LUMS
A Not-for-Profit University

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

TUTORIAL 2

Date: 27/09/2025.

Tutorial 2-1 10 marks

(Matrix–vector multiplication as a linear transformation)
Let $A \in \mathbb{R}^{m\times n}$ with columns $a_1, \ldots, a_n \in \mathbb{R}^m$. Define $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(x) = Ax$.

(a) Prove that T is linear and that for any $x \in \mathbb{R}^n$,

$$Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n.$$

(b) Compute Ax for

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}, \qquad x = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix},$$

and write the result explicitly as a linear combination of the columns of A.

(c) Prove that the set $\{Ax : x \in \mathbb{R}^n\}$ is a subspace of $\mathbb{R}^m$. (Name this subspace.)

(d) Briefly (2–3 sentences) explain how this shows that predictions of any linear model $y = Ax$ are constrained to lie in the span of the feature columns, and why this matters in machine learning.

Tutorial 2-2 10 marks

(Subspaces arising from a matrix; row and column spaces)
Let $A \in \mathbb{R}^{m \times n}$.

(a) Prove that $\mathrm{Col}(A) \subseteq \mathbb{R}^m$ and $\mathrm{Row}(A) \subseteq \mathbb{R}^n$ are subspaces.

(b) Prove that $\dim(\mathrm{Col}(A)) = \dim(\mathrm{Row}(A))$. (You may use basic facts about row reduction.)

(c) For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix},$$

find bases for $\mathrm{Col}(A)$ and $\mathrm{Row}(A)$ and compute their dimensions.

(d) Explain (2–3 sentences) why the dimension of the column space reflects the number of *independent directions* available from the features in a dataset.

Tutorial 2-3 10 marks

(Trace, Frobenius norm, and data dimensionality)

Let $X \in \mathbb{R}^{d \times n}$ be a data matrix whose columns are n samples in $\mathbb{R}^d$.

(a) Prove that $\|X\|_F^2 = \text{trace}(X^T X)$, where $\|X\|_F$ is the Frobenius norm.

(b) Show that if $\text{rank}(X) = r$ then the set of all data vectors (columns of X) lies in an r-dimensional subspace of $\mathbb{R}^d$.

(c) Compute $\|X\|_F$ and the rank of

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}.$$

(d) In 2–3 sentences, interpret $\|X\|_F$ and $\text{rank}(X)$ as measures of magnitude and intrinsic dimensionality of a dataset.

Tutorial 2-4 10 marks

(Gram construction and similarity of feature vectors)

Let vectors $v_1, \ldots, v_k \in \mathbb{R}^n$ and form the matrix $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{n \times k}$. Define the Gram matrix $G = V^T V$.

(a) Prove that $G_{ij} = v_i^T v_j$ for all i, j.

(b) Prove that $\text{rank}(G) = \dim(\text{span}\{v_1, \ldots, v_k\})$.

(c) Compute G and its rank for

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

(d) In 2–3 sentences explain how the Gram matrix encodes geometric similarity between feature vectors and why its rank reveals the number of independent feature directions.

Tutorial 2-5 10 marks

k-Nearest Neighbours (kNN) — matrix form

Consider the following small training set in $\mathbb{R}^2$ with 5 points, their labels, and a query point:

$$X = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 2 & 2 \\ 3 & 1 \end{bmatrix}, \qquad y = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \\ +1 \end{bmatrix}, \qquad z = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

(a) Write a compact matrix–vector expression for the vector of squared Euclidean distances

$$d^2 = \begin{bmatrix} \|x_1 - z\|^2 \\ \vdots \\ \|x_5 - z\|^2 \end{bmatrix} \in \mathbb{R}^5$$

using only X and z (no explicit loops). Then compute d^2 numerically for the dataset above, showing the algebraic steps.

(b) Using unweighted k-NN with Euclidean distance and $k = 3$, find the predicted label for z. Show which indices are the 3 nearest neighbors and perform the majority vote. Specify a deterministic tie-break rule if ties occur.

— End of Tutorial —